

Observability Is Restorability

Restorable Distinctions and the Geometry of Perceptual Access

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Abstract

Standard theories of observation treat the observer as a receiver: a system that acquires, records, or represents information from an external world. This paper proposes an alternative foundation. A distinction that cannot be restored after perturbation is operationally indistinguishable from a distinction that never existed. Observation therefore depends not on reception but on *restorability*. We introduce a pre-theoretic notion of *stable observability* and prove that it coincides exactly with finite-cost restorability under admissible perturbation—making the title thesis a theorem rather than a definition. An observer is then defined as a repair process whose distinction space is induced by its repair repertoire, and the algebra of repair spaces is developed: monotonicity, repair equivalence, subadditivity, the interference functional, the repair pseudometric, and the Reachability–Restorability inclusion. The boundary of the observer-relative admissibility manifold A_O is characterised as the divergence locus of minimal repair cost C_R —the Repair Boundary Theorem, grounded in a lemma on lower semi-continuity of repair cost. Applications are developed for blindsight and perceptual filling-in, quantum measurement as repair-or-collapse, and institutional observers. The framework unifies distinction theory, the admissibility program, and repair-theoretic accounts of memory and understanding into a single foundational identification.

1. Observation as a Problem

Standard theories of observation begin with reception. An observer receives signals, records measurements, forms representations, or updates beliefs [6, 16]. In every case observation is treated as the acquisition of information from an external world. The observer is essentially passive: a container into which the world deposits its impressions [5].

This paper begins from a different premise. A distinction that cannot be restored after perturbation is operationally indistinguishable from a distinction that never existed. Observation therefore depends not on reception but on restorability. What an observer can observe is precisely what it can restore.

We propose that observability and restorability are equivalent notions. An observer is not fundamentally a receiver of information but a repair process acting on a substrate of distinctions. Perception, memory, measurement, and scientific inquiry become special cases of distinction restoration under perturbation.

The central identification is:

$$\text{Obs}(O) = \text{Rest}(O), \tag{1}$$

where $\text{Obs}(O)$ is the set of distinctions observable by O and $\text{Rest}(O)$ is the set of distinctions restorable by O . Everything else in this paper is a consequence of taking (1) seriously.

The argument proceeds as follows. Section 2 establishes the mathematical substrate: state spaces, perturbations, and distinctions. Section 3 defines repair operators, repair paths, and repair cost. Section 4 introduces restorable distinctions. Section 5 re-defines the observer as a repair process whose distinction space is induced by its repair repertoire. Section 6 studies shared perceptual worlds as intersections of repair repertoires. Section 7 introduces the pre-theoretic notion of stable observability and proves that it coincides with restorability, making (1) a theorem rather than a stipulation. Section 8 develops the internal algebra of repair spaces: monotonicity, equivalence, subadditivity, the interference functional, the repair metric, and the reachability–restorability inclusion. Section ?? states and proves the Repair Boundary Theorem, supported by a lemma on lower semi-continuity of repair cost. Sections 9–11 develop applications. Section 12 draws consequences and identifies open problems.

2. Substrates, Perturbations, and Distinctions

Definition 2.1 (Substrate). A *substrate* is a triple $S = (X_S, \mathcal{P}, \mathcal{D})$ where:

- X_S is a topological space of *states*;
- $\mathcal{P}$ is a semigroup of *perturbation operators* $p : X_S \rightarrow X_S$ acting continuously on X_S , closed under composition, with identity $\text{id} \in \mathcal{P}$;
- $\mathcal{D} \subseteq X_S \times X_S$ is a symmetric, irreflexive *distinction relation*: $(x, y) \in \mathcal{D}$ means states x and y are discriminable.

Throughout this paper we impose two standing axioms on every substrate.

Axiom 2.2 (Locally compact Hausdorff substrate). The state space X_S is a locally compact Hausdorff topological space.

Axiom 2.3 (Closed distinction relation). The distinction relation $\mathcal{D}$ is closed in the product topology on $X_S \times X_S$.

Axiom 2.2 provides the compactness and separation needed for convergence arguments throughout. Axiom 2.3 ensures that limit states of sequences of discriminable pairs remain discriminable: if $(x_n, y_n) \in \mathcal{D}$ and $(x_n, y_n) \rightarrow (x, y)$, then $(x, y) \in \mathcal{D}$. Without this, the refinement step in Theorem 7.2 can fail. Both axioms hold in all concrete examples considered in this paper.

Remark 2.4. The distinction relation $\mathcal{D}$ need not be transitive and should not in general be an equivalence. Discrimination is a local, pair-wise affair. Two states may be mutually discriminable from a third without being mutually discriminable from each other.

Definition 2.5 (Admissible perturbation). A perturbation $p \in \mathcal{P}$ is *admissible relative to distinction* $d = (x, y)$ if p maps at least one of x, y to a state from which d might fail to hold — formally, if $(p(x), y) \notin \mathcal{D}$ or $(x, p(y)) \notin \mathcal{D}$. The set of admissible perturbations for d is denoted $\mathcal{P}_d$.

Definition 2.6 (Distinction survival). Distinction $d = (x, y)$ *survives perturbation* p if $(p(x), p(y)) \in \mathcal{D}$. Otherwise p *erases* d .

The central concern of this paper is what happens after erasure.

3. Repair Operators and Repair Cost

Definition 3.1 (Repair action). Given a substrate S , a *repair action* is a continuous map $r : X_S \rightarrow X_S$ belonging to a designated set $\mathcal{R} \subseteq C(X_S, X_S)$ of available interventions, where $C(X_S, X_S)$ carries the compact-open topology. The set $\mathcal{R}$ is the *repair repertoire* of the process under consideration. The study of controllable interventions on state spaces is treated systematically in [12, 11, 10].

Axiom 3.2 (Compact repair repertoire). The repair repertoire $\mathcal{R}$ is a sequentially compact subset of $C(X_S, X_S)$ under the compact-open topology. Consequently $\mathcal{R}^n$, the product space of repair paths of fixed length n , is compact by Tychonoff's theorem.

Definition 3.3 (Repair path). A *repair path* for distinction $d = (x_0, y_0)$ after perturbation p is a finite sequence of repair actions $\gamma = (r_1, r_2, \dots, r_n)$ such that, writing $x_k = r_k(x_{k-1})$ and $y_k = r_k(y_{k-1})$, we have $(x_n, y_n) \in \mathcal{D}$. The set of all repair paths of length at most N is denoted $\Gamma_d^{p,N}$; the union $\Gamma_d^p = \bigcup_{N \geq 1} \Gamma_d^{p,N}$.

Definition 3.4 (Repair cost). Let $c : X_S \rightarrow \mathbb{R}_{\geq 0}$ be a lower semi-continuous *local cost density* representing the resource expenditure per unit state occupancy. Typical interpretations include metabolic energy ($c(x) = \text{ATP consumption rate}$), computational effort ($c(x) = \text{clock cycles per step}$), or temporal cost ($c(x) = 1$ uniformly). The cost of a repair path $\gamma = (r_1, \dots, r_n)$ is the *discrete path sum*

$$\text{cost}(\gamma, x_0) = \sum_{k=1}^n c(x_{k-1}),$$

where $x_k = r_k(x_{k-1})$. When repair actions are parameterised as continuous flows $r_k = \Phi_{t_k}$ for $t_k > 0$, this sum extends naturally to the line integral $\int_{\gamma} c \, ds$; both formulations are used below according to context.

Definition 3.5 (Minimal repair cost). The *minimal repair cost* of distinction d after perturbation p , relative to repair repertoire $\mathcal{R}$, is

$$C_R(d, p) = \inf_{\gamma \in \Gamma_d^p} \text{cost}(\gamma, p(x_0)),$$

setting $C_R(d, p) = +\infty$ when $\Gamma_d^p = \emptyset$.

Definition 3.6 (Worst-case and probabilistic repair cost). The *worst-case repair*

cost of distinction d is

$$C_R(d) = \sup_{p \in \mathcal{D}_d} C_R(d, p).$$

This is the operationally severe notion: a single catastrophic perturbation with $C_R(d, p) = +\infty$ renders d non-restorable in the absolute sense.

For biological and institutional observers, a softer notion is often more appropriate. Given a probability measure μ on $\mathcal{D}_d$, the *probabilistic repair cost* is

$$C_R^\mu(d) = \int_{\mathcal{D}_d} C_R(d, p) d\mu(p).$$

Absolute restorability $\text{Rest}_\infty(\mathcal{R})$ uses C_R ; probabilistic restorability $\text{Rest}_\mu(\mathcal{R})$ uses C_R^μ . The former is the object of this paper unless otherwise stated.

4. Restorable Distinctions

Definition 4.1 (Restorability). A distinction $d \in \mathcal{D}$ is *restorable by repair repertoire* $\mathcal{R}$ if $C_R(d) < +\infty$ — that is, if every admissible perturbation admits a finite-cost repair path. This is *finite-step restorability*; the set is

$$\text{Rest}_f(\mathcal{R}) = \{d \in \mathcal{D} \mid C_R(d) < +\infty\}.$$

The finite-step condition requires a repair path of bounded length. A weaker but mathematically significant notion permits repair in the limit of infinitely many steps.

Definition 4.2 (Limit restorability). A distinction $d \in \mathcal{D}$ is *limit-restorable by* $\mathcal{R}$ if for every admissible perturbation $p \in \mathcal{D}_d$ and every $\varepsilon > 0$ there exists a (possibly infinite) sequence of repair actions $(r_1, r_2, \dots)$ such that the partial orbit (x_k, y_k) satisfies

$$\inf_{(u,v) \in \mathcal{D}} [\|x_k - u\| + \|y_k - v\|] \rightarrow 0 \quad \text{as } k \rightarrow \infty,$$

with $\sum_{k=1}^{\infty} c(x_{k-1}) < +\infty$. The set of limit-restorable distinctions is $\text{Rest}_\omega(\mathcal{R})$.

Proposition 4.3 (Inclusion of restorability classes). $\text{Rest}_f(\mathcal{R}) \subseteq \text{Rest}_\omega(\mathcal{R})$. The inclusion is in general strict.

Proof. Any finite repair path is a finite sequence; the partial orbit reaches $\mathcal{D}$

in finitely many steps, so the infimum condition holds in the limit. Strictness: consider a substrate in which every perturbation of d can be countered by successively halving the remaining distance to $\mathcal{D}$, with costs $1/2^k$; the total cost $\sum 1/2^k = 1$ is finite, but no single finite path lands in the closed set $\mathcal{D}$ unless the limit state is taken. Under Axiom 2.3 the limit is in $\mathcal{D}$, confirming limit-restorability; but if no path of fixed finite length suffices, the distinction is in $\text{Rest}_\omega \setminus \text{Rest}_f$. $\square$

Remark 4.4. The two classes separate different physical regimes. Biological repair (neural recovery, wound healing) and mathematical proof search aspire to Rest_f : a bounded sequence of discrete interventions must suffice. Iterative scientific refinement, asymptotic stabilisation of institutions, and language reconstruction may be modelled more faithfully by Rest_ω : convergence is guaranteed but no finite stopping point is specified. Courts occupy an intermediate position, tolerating appellate iteration but demanding termination. The Stable Observability Theorem (Theorem 7.2) is a theorem about Rest_f ; a parallel theorem for Rest_ω under weaker topological assumptions is an open problem (Problem 1 in Section 12).

Throughout Sections 5–??, $\text{Rest}(\mathcal{R})$ denotes $\text{Rest}_f(\mathcal{R})$ unless otherwise stated.

Proposition 4.5 (Restorability is closed under finite composition). *If $d_1, d_2 \in \text{Rest}(\mathcal{R})$ and the restoration of d_2 does not interfere with the restored state of d_1 (formally: the repair paths commute on the relevant component of X_S), then the compound distinction $d_1 \wedge d_2$ is also in $\text{Rest}(\mathcal{R})$.*

Proof. Let $\gamma_1 \in \Gamma_{d_1}^p$ and $\gamma_2 \in \Gamma_{d_2}^p$ be finite-cost repair paths. Under the commutativity assumption, $\gamma_1 \cdot \gamma_2$ (concatenation) restores both distinctions with cost at most $\text{cost}(\gamma_1) + \text{cost}(\gamma_2) < +\infty$. $\square$

Remark 4.6. The commutativity assumption in Proposition 4.5 is non-trivial and fails in general. Section 9 exhibits cases where restoring one distinction actively interferes with another, producing *repair conflict*.

5. Observers as Repair Processes

The standard move is to define an observer as a pair $(O, \mathcal{D}_O)$ of a process and a set of distinctions it tracks [1, 2]. We invert this dependency.

Definition 5.1 (Observer). An *observer* is a repair process $O = (S, \mathcal{R}_O)$ consisting of a substrate S and a repair repertoire $\mathcal{R}_O$. The *observation space* of O is the induced set

$$\mathcal{D}_O = \text{Rest}(\mathcal{R}_O).$$

The distinction space is not given in advance and then repaired. It is the repair repertoire that determines which distinctions exist for this observer. An observer with a larger repertoire inhabits a richer distinction space, not because it receives more information but because it can restore more.

Definition 5.2 (Observability). A distinction d is *observable by* O if and only if $d \in \text{Rest}(\mathcal{R}_O)$. In symbols:

$$\text{Obs}(O) = \text{Rest}(\mathcal{R}_O) = \mathcal{D}_O.$$

This is the formal content of (1). Observability just is restorability.

6. Perceptual Worlds and Observer Agreement

Definition 6.1 (Shared perceptual world). Two observers O_1 and O_2 *share a perceptual world* to the extent that their observation spaces overlap:

$$W(O_1, O_2) = \text{Obs}(O_1) \cap \text{Obs}(O_2) = \text{Rest}(\mathcal{R}_{O_1}) \cap \text{Rest}(\mathcal{R}_{O_2}).$$

Theorem 6.2 (Shared perception is shared repair). *Two observers share a perceptual world if and only if they share repair capacity over that world:*

$$W(O_1, O_2) = \text{Rest}(\mathcal{R}_{O_1}) \cap \text{Rest}(\mathcal{R}_{O_2}).$$

In particular, two observers with disjoint repair repertoires inhabit disjoint perceptual worlds even when embedded in the same physical substrate.

Proof. Immediate from Definition 5 and the definition of $W(O_1, O_2)$. □

Corollary 6.3 (Observer agreement). *Two observers agree on distinction d if and only if $d \in W(O_1, O_2)$, i.e., both can restore d after any admissible perturbation. Agreement on signals is neither necessary nor sufficient; what matters is agreement on repairability.*

Example 6.4 (Species-relative perception). A bat and a human occupy the same physical environment. Their repair repertoires differ substantially in the echoic dimension. Theorem 6.2 predicts that the spatial distinctions available to the bat via echolocation are not observable by the human, not because the signals are absent but because the human lacks repair operators acting on that channel.

7. Stable Observability

The identification $\text{Obs}(O) = \text{Rest}(O)$ as stated in Section 5 is definitional: observability was *defined* as restorability. A genuine theorem requires a pre-theoretic notion of observability that is independent of the repair apparatus, so that the equivalence can be established rather than assumed.

We supply one now.

Definition 7.1 (Stable observability). Fix an observer $O = (S, \mathcal{R}_O)$ and a threshold $\varepsilon > 0$. A distinction $d = (x, y)$ is *stably observable* by O if for every $\delta > 0$ and every admissible perturbation $p \in \mathcal{P}_d$, there exists a finite sequence of repair actions $(r_1, \dots, r_n) \in \mathcal{R}_O^*$ such that the resulting states (x', y') satisfy

$$\inf_{(u,v) \in \mathcal{D}} [\|x' - u\| + \|y' - v\|] < \delta,$$

and the total cost of the sequence is bounded independently of δ . The set of stably observable distinctions is denoted $\text{Obs}_{\text{stab}}(O)$.

Informally: d is stably observable if, no matter how it is perturbed, O can return to a state of discrimination within finite cost. The definition is pre-theoretic in the sense that it makes no reference to repair paths as formal objects; it speaks only of the capacity to discriminate after perturbation.

Theorem 7.2 (Stable Observability Theorem). *Under Axioms 2.2, 2.3, and 3.2, and assuming $c : X_S \rightarrow \mathbb{R}_{\geq 0}$ is lower semi-continuous,*

$$\text{Obs}_{\text{stab}}(O) = \text{Rest}(\mathcal{R}_O).$$

Proof. ($\supseteq$) Suppose $d \in \text{Rest}(\mathcal{R}_O)$. Then $C_R(d) < +\infty$, so every admissible perturbation admits a finite-cost repair path $\gamma^* \in \Gamma_d^p$ that lands in $\mathcal{D}$. The dis-

crimination condition in the definition of stable observability is satisfied with $\delta = 0$ (strictly inside $\mathcal{D}$), and the cost bound $C = C_R(d)$ is independent of δ . Thus $d \in \text{Obs}_{\text{stab}}(O)$.

($\subseteq$) Suppose $d = (x_0, y_0) \in \text{Obs}_{\text{stab}}(O)$. Fix an admissible perturbation $p \in \mathcal{D}_d$. By stable observability, for each $m \geq 1$ there exists a finite repair sequence $\gamma_m = (r_{1,m}, \dots, r_{n_m,m})$ of total cost $\text{cost}(\gamma_m) \leq C_p < +\infty$ (with C_p independent of m) such that the terminal states (x'_m, y'_m) satisfy

$$\inf_{(u,v) \in \mathcal{D}} [\|x'_m - u\| + \|y'_m - v\|] < \frac{1}{m}.$$

Bounding the path length. Since c is lower semi-continuous and positive, and X_S is locally compact, there exists a lower bound $c_0 > 0$ on any compact neighbourhood of the orbit. Each repair step contributes at least c_0 to the cost, so $n_m \leq C_p/c_0 =: N$ uniformly in m . We may therefore assume all paths have the same length $n \leq N$ (padding with the identity if needed).

Extracting a convergent subsequence. By Axiom 3.2, $\mathcal{R}_O$ is sequentially compact under the compact-open topology, so $\mathcal{R}_O^n$ is compact by Tychonoff. The sequence of tuples $(\gamma_m)_{m \geq 1}$ in $\mathcal{R}_O^n$ admits a convergent subsequence $(\gamma_{m_j}) \rightarrow \gamma^* = (r_1^*, \dots, r_n^*)$. By joint continuity of evaluation in the compact-open topology on locally compact Hausdorff spaces (Axiom 2.2), the terminal states converge: $(x'_{m_j}, y'_{m_j}) \rightarrow (x'^*, y'^*)$.

Landing in $\mathcal{D}$. For each j , the terminal states satisfy $\inf_{(u,v) \in \mathcal{D}} [\|x'_{m_j} - u\| + \|y'_{m_j} - v\|] < 1/m_j$, so (x'_{m_j}, y'_{m_j}) is within $1/m_j$ of $\mathcal{D}$. In the limit, $(x'^*, y'^*) \in \overline{\mathcal{D}} = \mathcal{D}$ by Axiom 2.3. Therefore $\gamma^* \in \Gamma_d^p$ is a valid repair path.

Cost bound. The total cost functional $\text{cost}(\gamma, x) = \sum_{k=1}^n c(x_{k-1})$ is a finite sum of lower semi-continuous functions under the compact-open topology and is therefore lower semi-continuous itself. Thus

$$\text{cost}(\gamma^*) \leq \liminf_j \text{cost}(\gamma_{m_j}) \leq C_p < +\infty.$$

Since this holds for every admissible perturbation p , $C_R(d) \leq C_p < +\infty$, so $d \in \text{Rest}(\mathcal{R}_O)$. $\square$

Remark 7.3. Theorem 7.2 is the paper's load-bearing result. It establishes that the title thesis—observability is restorability—is not a definitional convenience but

a substantive claim: the pre-theoretic capacity to maintain discrimination under perturbation is exactly captured by the formal notion of finite-cost repair. Every subsequent theorem is a consequence of taking this equivalence seriously.

8. The Algebra of Repair Spaces

We now develop the internal mathematics of repair spaces before connecting them to the admissibility manifold. The results of this section do not depend on Theorem 7.2 and hold for any repair repertoire.

8.1. Monotonicity

Theorem 8.1 (Repair Monotonicity). *Let $\mathcal{R}_1 \subseteq \mathcal{R}_2$ be repair repertoires acting on the same substrate S . Then*

$$\text{Rest}(\mathcal{R}_1) \subseteq \text{Rest}(\mathcal{R}_2).$$

Proof. If $d \in \text{Rest}(\mathcal{R}_1)$, every admissible perturbation of d admits a finite-cost repair path composed entirely of elements of $\mathcal{R}_1$. Since $\mathcal{R}_1 \subseteq \mathcal{R}_2$, every such path is available in $\mathcal{R}_2$, so $C_R[\mathcal{R}_2](d) \leq C_R[\mathcal{R}_1](d) < +\infty$. Thus $d \in \text{Rest}(\mathcal{R}_2)$. $\square$

Corollary 8.2 (Learning enlarges worlds). *Whenever $\mathcal{R}_{O_1} \subseteq \mathcal{R}_{O_2}$,*

$$\text{Obs}(O_1) \subseteq \text{Obs}(O_2).$$

Acquiring new repair operators strictly enlarges the set of observable distinctions. This gives a precise sense in which learning enlarges the world an observer inhabits: not by receiving more signals, but by gaining more repair capacity.

8.2. Observer Equivalence

Definition 8.3 (Repair equivalence). *Observers O_1 and O_2 are repair-equivalent, written $O_1 \sim_R O_2$, if $\text{Rest}(\mathcal{R}_{O_1}) = \text{Rest}(\mathcal{R}_{O_2})$.*

Theorem 8.4 (Repair Equivalence Theorem). *Repair-equivalent observers are observationally indistinguishable: $O_1 \sim_R O_2$ implies $\text{Obs}(O_1) = \text{Obs}(O_2)$.*

Proof. Immediate from the identification $\text{Obs}(O) = \text{Rest}(\mathcal{R}_O)$. $\square$

Remark 8.5. Theorem 8.4 is a repair-theoretic analogue of bisimulation in process calculi. Two observers may have entirely different internal architectures and yet be observationally equivalent if their repair repertoires generate the same restorable set. This suggests that the natural equivalence class for observer identity is repair capacity, not physical constitution—a conclusion consonant with the autopoietic account of identity [3, 4] but grounded here in the repair structure rather than in biological self-production.

8.3. Subadditivity and the Interference Functional

Theorem 8.6 (Subadditivity of Repair Cost). *Suppose distinctions d_1 and d_2 admit commuting repair paths (as in Proposition 4.5). Then*

$$C_R(d_1 \wedge d_2) \leq C_R(d_1) + C_R(d_2).$$

Proof. Let $\gamma_1^* \in \Gamma_{d_1}^p$ and $\gamma_2^* \in \Gamma_{d_2}^p$ be minimal repair paths for each distinction separately. Under commutativity, the concatenation $\gamma_1^* \cdot \gamma_2^*$ is a valid repair path for $d_1 \wedge d_2$ with cost at most $\text{cost}(\gamma_1^*) + \text{cost}(\gamma_2^*)$. Taking infima on each side yields the stated inequality. $\square$

Definition 8.7 (Repair interference functional). For distinctions $d_1, d_2 \in \text{Rest}(\mathcal{R})$, define

$$I(d_1, d_2) = C_R(d_1 \wedge d_2) - C_R(d_1) - C_R(d_2).$$

By Theorem 8.6, $I(d_1, d_2) \leq 0$ when repair paths commute. Three regimes are distinguished:

- $I(d_1, d_2) > 0$: *repair conflict*. Restoring both distinctions jointly costs more than restoring each separately. This is the formal signature of perceptual trade-offs: attending to one distinction impedes maintenance of another.

- $I(d_1, d_2) = 0$: *repair independence*. The distinctions are orthogonal in repair space.
- $I(d_1, d_2) < 0$: *repair synergy*. Joint restoration is cheaper than independent restoration. This captures perceptual binding: some distinctions are more efficiently maintained together than apart, which may underlie Gestalt grouping effects and multi-modal perceptual integration.

When the restorable distinction space is finite or countable, the pairwise values $I(d_i, d_j)$ assemble into a matrix.

Definition 8.8 (Interference matrix). Let $\{d_1, \dots, d_n\} \subseteq \text{Rest}(\mathcal{R})$ be a finite collection of distinctions. The *interference matrix* is

$$\mathbf{I} = (I(d_i, d_j))_{i,j=1}^n \in \mathbb{R}^{n \times n}.$$

Proposition 8.9 (Spectral interpretation of the interference matrix). $\mathbf{I}$ is symmetric. Its spectrum $\text{spec}(\mathbf{I})$ carries the following interpretation:

1. Large positive eigenvalues: the corresponding eigenvectors identify repair bottleneck clusters—groups of distinctions that mutually impede each other’s restoration. These correspond to attentional capacity limits in biological observers and bandwidth constraints in institutional ones.
2. Large negative eigenvalues: the eigenvectors identify synergistic bundles—groups of distinctions that are substantially cheaper to restore jointly. These correspond to Gestalt perceptual groupings, co-regulated gene networks, and interdependent institutional procedures.
3. Near-zero eigenvalues: the associated distinctions are approximately independent in repair space—neither conflicting nor synergistic.

The Frobenius norm $\|\mathbf{I}\|_F$ quantifies the total interaction strength; $\|\mathbf{I}\|_F = 0$ iff all distinctions in the collection are repair-independent.

Proof. Symmetry: $I(d_i, d_j) = I(d_j, d_i)$ since $d_i \wedge d_j = d_j \wedge d_i$ and C_R is a function of the distinction alone. The spectral interpretations follow from standard spectral

geometry: the quadratic form $v^\top \mathbf{I} v$ for a unit vector v represents the weighted interference over the combination $\sum_i v_i d_i$, with positive values indicating net conflict and negative values indicating net synergy. $\square$

Remark 8.10. The interference matrix has the algebraic structure of a signed Gram matrix over the distinction space, and its spectral theory connects the repair framework to random matrix theory and the theory of signed networks. For an observer with a large restorable distinction space, the density of states of $\mathbf{I}$ describes the global interaction structure of perception. A repair bottleneck corresponds to a rank-one positive perturbation of $\mathbf{I}$; the eigenvalue gap around zero measures how close the observer is to a globally synergistic or globally conflicted regime. Ray Barman et al. [19] demonstrate empirically that interference among competing memories—rather than temporal decay alone—drives power-law forgetting in high-dimensional retrieval systems, and that vulnerability to interference is governed by *effective* rather than nominal dimensionality. The interference matrix provides a formal object in which these empirical phenomena have a natural home: the effective dimensionality of a retrieval space corresponds to the rank of $\mathbf{I}$, and the concentration of variance into few dimensions corresponds to a low-rank positive-semidefinite structure with a small number of large positive eigenvalues.

8.4. The Repair Pseudometric

Definition 8.11 (Repair pseudometric). For distinctions $d_i, d_j \in \text{Rest}(\mathcal{R})$, define

$$d_R(d_i, d_j) = \inf_{\gamma: d_i \rightsquigarrow d_j} \int_{\gamma} c(x) \, ds,$$

where the infimum is taken over all repair trajectories in X_S that transform the substrate state associated with d_i into one from which d_j is restorable. The construction is formally analogous to the Kantorovich–Wasserstein distance on probability measures [15], with repair trajectories playing the role of transport plans.

Proposition 8.12 (Repair pseudometric properties). *The function d_R satisfies:*

1. $d_R(d_i, d_i) = 0$ for all d_i ;
2. $d_R(d_i, d_k) \leq d_R(d_i, d_j) + d_R(d_j, d_k)$ (triangle inequality, by path concatenation);

3. If repair trajectories are reversible and c is symmetric under reversal, then $d_R(d_i, d_j) = d_R(d_j, d_i)$ and d_R is a genuine metric on $\text{Rest}(\mathcal{R})$.

Remark 8.13. Irreversible perturbations—cell death, archive destruction, data erasure—produce cases where $d_R(d_i, d_j) \neq d_R(d_j, d_i)$. The asymmetric pseudometric is therefore the primary object; the metric is a special case arising under reversibility. The directed structure of irreversible repair spaces may connect to the time-asymmetric picture developed in the RSVP framework.

8.5. Reachability and Restorability

The following theorem is the formal weld between the repair-observer framework and the admissibility program.

Theorem 8.14 (Reachability–Restorability Inclusion). *Let A_O be the observer-relative admissibility manifold (Definition of Section ??) and let $\text{Reach}(A_O)$ denote the set of distinctions reachable within A_O under the dynamics of S . Then*

$$\text{Rest}(\mathcal{R}_O) \subseteq \text{Reach}(A_O).$$

Every observable distinction lies within the observer’s reachable admissibility region.

Proof. Suppose $d \in \text{Rest}(\mathcal{R}_O)$. Then $C_R(d) < +\infty$, so there exists a finite-cost repair path γ that restores d from any admissible perturbation. This path lies within X_S and terminates in a state where d is discriminable. By definition, the set of states from which d remains restorable is contained in A_O (since those states are precisely those from which the observer can maintain its distinctions). The path γ therefore connects the perturbed state to a point in A_O , establishing that d is reachable within A_O . $\square$

Remark 8.15. The inclusion $\text{Rest}(\mathcal{R}_O) \subseteq \text{Reach}(A_O)$ is in general strict. Equality would require that every distinction reachable within the admissibility manifold is also restorable by the repair repertoire, which fails when the manifold contains states that are geometrically accessible but dynamically blocked. Whether equality holds under compactness and convexity assumptions is an open question (Problem 5 in Section 12).

8.6. Control–Geometry Duality

The Reachability–Restorability Inclusion reveals a structural duality running through the entire framework. On one side stands the *control* perspective: the repair repertoire $\mathcal{R}_O$ generates a dynamics on X_S , and questions of repair are questions of controllability. On the other side stands the *geometry* perspective: the restorable distinction space $\text{Rest}(\mathcal{R}_O)$ inherits a metric structure from repair cost, and questions of observation are questions of geometry.

Definition 8.16 (Control structure and induced geometry). The *control structure* of an observer is the triple $\mathfrak{C}(\mathcal{R}_O) = (X_S, \mathcal{R}_O, c)$: a state space, a set of admissible interventions, and a cost density. The *induced geometry* is the pair $\mathfrak{G}(\text{Rest}(\mathcal{R}_O)) = (\text{Rest}(\mathcal{R}_O), d_R)$: the restorable distinction space equipped with the repair pseudometric.

The duality takes the form of a correspondence:

Reachability in $\mathfrak{C}$	$\longleftrightarrow$	Existence of a repair trajectory
Restorability in $\mathfrak{G}$	$\longleftrightarrow$	Finite cost of a repair trajectory
Admissibility boundary ∂A_O	$\longleftrightarrow$	Divergence locus of C_R
Local controllability	$\longleftrightarrow$	Open geodesic balls in d_R
Topological controllability	$\longleftrightarrow$	Dense repair orbits in $\text{Rest}(\mathcal{R}_O)$

Theorem 8.17 (Control–Geometry Collapse). *Let $X_S \subseteq \mathbb{R}^m$ be compact and convex and let the repair repertoire $\mathcal{R}_O$ consist of continuous maps forming a locally state-controllable system on the interior of A_O . Then*

$$\text{Rest}(\mathcal{R}_O) = \text{Reach}(A_O).$$

Proof. ($\subseteq$) Theorem 8.14.

($\supseteq$) Suppose $d \in \text{Reach}(A_O)$. By local state-controllability, the reachable set from any interior point of A_O is open. Any continuous path in A_O from the perturbed state to a state where d is discriminable can therefore be covered by finitely many open controllable neighbourhoods (by compactness of A_O). Each

neighbourhood transition is realised by a finite sequence of repair operators in $\mathcal{R}_O$. The composite path has total cost bounded by $N \cdot M$ where N is the number of transitions and $M = \max_{x \in X_S} c(x) < +\infty$ (since c is lower semi-continuous on a compact space). Thus $C_R(d) \leq N \cdot M < +\infty$, so $d \in \text{Rest}(\mathcal{R}_O)$. $\square$

Remark 8.18. Theorem 8.17 is a resolution of Problem 5 under explicit hypotheses. The control-theoretic condition (local state-controllability) is the precise content of the intuition that “every geometrically accessible state is dynamically reachable.” When $\mathcal{R}_O$ generates a dense subgroup of the homeomorphism group of X_S , controllability holds automatically and the equality follows. The Repair Boundary Theorem is then a corollary of the duality: the boundary ∂A_O is simultaneously the set of states from which the geometry runs out (repair cost diverges) and the set of states from which the control system can no longer steer.

We characterise the boundary of the observer-relative admissibility manifold in terms of diverging repair cost. Theorem 8.14 established that every restorable distinction lies within the reachable admissibility region; the Repair Boundary Theorem pinpoints where that region ends. It requires a preparatory lemma.

Lemma 8.19 (Lower semi-continuity of repair cost). *Let X_S be locally compact (Axiom 2.2), let the local cost density $c : X_S \rightarrow \mathbb{R}_{\geq 0}$ be lower semi-continuous [14], and let the admissible repair paths form a family closed under uniform limits on compact intervals. Then the minimal repair cost*

$$C_R(d, x) = \inf_{\gamma \in \Gamma_d(x)} \int_{\gamma} c \, ds$$

is lower semi-continuous in x .

Proof. Let (x_n) be a sequence in X_S converging to x . We show $\liminf_{n \rightarrow \infty} C_R(d, x_n) \geq C_R(d, x)$.

For each n , let $\gamma_n \in \Gamma_d(x_n)$ be a near-optimal repair path with $\text{cost}(\gamma_n) \leq C_R(d, x_n) + 1/n$. Restricting to a compact neighbourhood K of x and using local compactness, the paths (γ_n) are equibounded in cost. By the assumed closure under uniform limits on compact intervals, a subsequence (γ_{n_k}) converges uniformly to a limit path $\gamma^* \in \Gamma_d(x)$.

By lower semi-continuity of c and Fatou’s lemma applied to the path integrals,

$$\int_{\gamma^*} c \, ds \leq \liminf_{k \rightarrow \infty} \int_{\gamma_{n_k}} c \, ds \leq \liminf_{n \rightarrow \infty} C_R(d, x_n).$$

Since $\gamma^* \in \Gamma_d(x)$, we have $C_R(d, x) \leq \int_{\gamma^*} c \, ds$, completing the proof. $\square$

Definition 8.20 (Observer-relative admissibility manifold). Let $O = (S, \mathcal{R}_O)$ be an observer. The *observer-relative admissibility manifold* A_O is the closure of the set of states from which all distinctions in $\mathcal{D}_O$ remain restorable:

$$A_O = \overline{\{x \in X_S \mid \forall d \in \mathcal{D}_O, C_R(d) < +\infty \text{ from } x\}}.$$

Theorem 8.21 (Repair Boundary Theorem). *A state x lies on the boundary of the observer-relative admissibility manifold if and only if the minimal repair cost of at least one distinction in $\mathcal{D}_O$ diverges as the system approaches x :*

$$x \in \partial A_O \iff \exists d \in \mathcal{D}_O : C_R(d) \rightarrow +\infty \text{ as the system state approaches } x.$$

Proof. ($\Rightarrow$) Suppose $x \in \partial A_O$. Then x is a limit point of A_O and also of its complement. In any neighbourhood $U \ni x$ there exist states from which some distinction $d \in \mathcal{D}_O$ is not finitely restorable. By Lemma 8.19, $C_R(d, \cdot)$ is lower semi-continuous; hence $C_R(d)$ must become unbounded along any sequence approaching x from outside A_O , establishing $C_R(d) \rightarrow +\infty$.

($\Leftarrow$) If $C_R(d) \rightarrow +\infty$ as the state approaches x , then in any neighbourhood of x there are states from which d is not finitely restorable. Hence $x \notin \text{int}(A_O)$, so $x \in \partial A_O$. $\square$

Remark 8.22. Theorem 8.21 identifies the admissibility boundary not as an externally imposed constraint but as an emergent consequence of the repair structure. The boundary is where the observer's world ends, in the precise sense that no further repair is possible.

Corollary 8.23 (Admissibility as repair capacity). *The interior of A_O is exactly the set of states from which the observer can maintain all its distinctions under all admissible perturbations. Moving toward ∂A_O is operationally equivalent to losing observational capacity.*

9. Blindsight and Perceptual Filling-In

The repair-observer framework generates a clean account of *blindsight*: the phenomenon in which subjects with cortical damage respond discriminatively to visual stimuli they cannot consciously report [7]. The division of visual processing into action-guiding and perception-reporting streams [8] maps directly onto the sub-report/report-level restorability split defined below.

In the standard receptive model, blindsight is paradoxical: how can information be used if it is not received at the reporting level?

Under the repair model the paradox dissolves. Let the observer O decompose into repair sub-processes operating at different substrate levels: $O = O_{\text{sub}} \cup O_{\text{rep}}$, where O_{sub} acts below the reporting interface and O_{rep} acts at the interface itself.

Definition 9.1 (Sub-report restorability). A distinction d is *sub-report restorable* if $d \in \text{Rest}(\mathcal{R}_{O_{\text{sub}}})$ but $d \notin \text{Rest}(\mathcal{R}_{O_{\text{rep}}})$.

Blindsight is then the condition in which the distinction is sub-report restorable: the sub-cortical repair process maintains the distinction, but the cortical repair process (whose outputs constitute verbal report) does not have access to a repair path. The subject *observes* in the sub-report sense while failing to *observe* in the report sense.

Remark 9.2. This account predicts that blindsight subjects should show above-chance performance on forced-choice tasks (sub-report restoration succeeds) while failing to initiate voluntary report (report-level restoration is unavailable). This is the observed pattern.

Perceptual filling-in admits a parallel treatment. The visual system restores distinctions at the boundary of the blind spot not by receiving signals from that region but by extending repair trajectories from surrounding states. The filled-in percept is a repair product, not a received datum.

10. Quantum Measurement as Repair or Collapse

Standard quantum measurement theory distinguishes pre-measurement superposition from post-measurement eigenstate. The *measurement problem* asks what physical process constitutes the collapse.

The repair-observer framework does not solve the measurement problem but reframes it productively. The present account is complementary to relational quantum mechanics [17], which treats measurement outcomes as relative to observers, and to the decoherence program [18], which identifies the mechanism by which classical pointer states become stable. Under the repair model, decoherence is the process by which the apparatus acquires repair operators capable of acting on classical pointer distinctions.

Let the substrate S be a quantum system with Hilbert space $\mathcal{H}$ and let the distinction relation $\mathcal{D}$ be defined on density operators: $(\rho_1, \rho_2) \in \mathcal{D}$ iff $\|\rho_1 - \rho_2\|_1 > \varepsilon$ for some threshold $\varepsilon > 0$.

A perturbation p that places the system in a superposition $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ erases the distinction between $|0\rangle$ and $|1\rangle$ at the classical level (neither eigenstate is discriminated from the other in $\mathcal{H}$ under p).

Measurement is then a repair event: either the apparatus-observer O_{app} possesses a repair operator that drives the state to a definite eigenstate (collapse as successful restoration), or it does not (the distinction remains unrestorable at the classical interface).

Proposition 10.1 (Measurement as repair completion). *A measurement interaction constitutes observation of distinction $d = (|0\rangle, |1\rangle)$ by apparatus O_{app} if and only if O_{app} possesses a repair path that takes the post-perturbation superposition to a state in which d is discriminable at the pointer basis.*

Remark 10.2. This connects to the quadratic gravity interface picture developed elsewhere in the Flyxion program: the interface at which quantum distinctions become classically available is the interface at which repair operators acting on the classical substrate can restore those distinctions. Decoherence is the mechanism by which this interface becomes accessible.

11. Institutional Observers

Individual biological observers are a special case. The framework is equally applicable to any process with a repair repertoire.

Definition 11.1 (Institutional observer). *An institutional observer is an observer $O_{\text{inst}} = (S, \mathcal{R}_{\text{inst}})$ in which the substrate S is a social or informational system and the repair repertoire $\mathcal{R}_{\text{inst}}$ consists of codified institutional procedures.*

Example 11.2 (Legal systems). A court is an institutional observer whose substrate is the space of legally characterised states and whose repair repertoire consists of evidentiary and procedural rules. A distinction d between *legal* and *illegal* conduct is observable by the court if and only if the evidentiary procedures can restore that distinction after the perturbation of competing characterisations.

When courts collapse or procedures are suspended, the distinction does not become *false*: it becomes *unrestorable*. This is the precise sense in which institutional decay is observational decay.

Example 11.3 (Archives and libraries). An archive restores distinctions across time by maintaining repair paths from present queries to past states. A destroyed archive does not falsify historical distinctions; it removes the repair operators by which those distinctions could be recovered.

Example 11.4 (Scientific laboratories). A laboratory is an institutional observer specialised in maintaining distinctions under repeated, intersubjectively verified perturbation. Scientific progress is the expansion of $\text{Rest}(\mathcal{R}_{\text{lab}})$: the set of distinctions that the collective repair apparatus of science can maintain. A scientific *revolution* (in Kuhn's sense) is a restructuring of $\mathcal{R}_{\text{lab}}$ that changes which distinctions are restorable, often making previously central distinctions unrestorable while opening new regions of restoration.

Theorem 11.5 (Paradigm Shift as Repair-Repertoire Discontinuity). *Let $\mathcal{R}_{\text{old}}$ and $\mathcal{R}_{\text{new}}$ be two repair repertoires representing a scientific community before and after a paradigm shift. If the shift is non-trivial, meaning $\mathcal{R}_{\text{old}} \not\subseteq \mathcal{R}_{\text{new}}$ and $\mathcal{R}_{\text{new}} \not\subseteq \mathcal{R}_{\text{old}}$, then the restorable distinction sets are incomparable:*

$$\text{Rest}(\mathcal{R}_{\text{old}}) \not\subseteq \text{Rest}(\mathcal{R}_{\text{new}}) \quad \text{and} \quad \text{Rest}(\mathcal{R}_{\text{new}}) \not\subseteq \text{Rest}(\mathcal{R}_{\text{old}}).$$

In particular, the symmetric difference

$$\text{Rest}(\mathcal{R}_{\text{old}}) \triangle \text{Rest}(\mathcal{R}_{\text{new}}) \neq \emptyset$$

contains lost distinctions $\text{Rest}(\mathcal{R}_{\text{old}}) \setminus \text{Rest}(\mathcal{R}_{\text{new}})$ and gained distinctions $\text{Rest}(\mathcal{R}_{\text{new}}) \setminus \text{Rest}(\mathcal{R}_{\text{old}})$.

Proof. By contraposition of Theorem 8.1: if $\mathcal{R}_{\text{old}} \not\subseteq \mathcal{R}_{\text{new}}$, there exists $r \in \mathcal{R}_{\text{old}} \setminus \mathcal{R}_{\text{new}}$. Any distinction d whose only repair paths under admissible perturbations pass through r is in $\text{Rest}(\mathcal{R}_{\text{old}}) \setminus \text{Rest}(\mathcal{R}_{\text{new}})$; such distinctions exist generically

when r is not compositionally redundant. The symmetric argument yields gains. $\square$

Remark 11.6. Theorem 11.5 formalises Kuhnian incommensurability [9] as a repair-space statement. Lost distinctions are not *refuted*; they become *unrestorable* in the new paradigm. This is precisely the claim that different paradigms make different kinds of mistakes visible and invisible, not merely different claims about the same visible space. The interference matrix $\mathbf{I}$ of Section 8 will in general have a different spectrum under $\mathcal{R}_{\text{old}}$ and $\mathcal{R}_{\text{new}}$, providing a quantitative signature of the shift. This connects the theorem to the distinction-reorganization papers in the broader program.

Definition 11.7 (Social reality as institutional restorability). An institutional distinction d is *socially real* relative to institutional observer O_{inst} if and only if $d \in \text{Rest}(\mathcal{R}_{\text{inst}})$.

Proposition 11.8 (Institutional collapse shrinks the real). *Let O_{inst} undergo a perturbation that reduces its repair repertoire from $\mathcal{R}$ to $\mathcal{R}' \subsetneq \mathcal{R}$. Then*

$$\text{Rest}(\mathcal{R}') \subseteq \text{Rest}(\mathcal{R}),$$

and the set of socially real distinctions is (weakly) smaller after the collapse. Distinctions in $\text{Rest}(\mathcal{R}) \setminus \text{Rest}(\mathcal{R}')$ do not become false; they become unrestorable.

Proof. By Theorem 8.1 applied contravariantly: $\mathcal{R}' \subseteq \mathcal{R}$ implies $\text{Rest}(\mathcal{R}') \subseteq \text{Rest}(\mathcal{R})$. Any distinction in the difference set was formerly restorable by procedures no longer available; it is now unrestorable, hence socially unreal by Definition 11.7. $\square$

12. Consequences and Open Problems

Taking $\text{Obs}(O) = \text{Rest}(O)$ seriously reorganises several independent lines of inquiry into a single structure.

Memory. Memory is not storage of representations but maintenance of repair trajectories. To remember a distinction is to retain a repair path by which it can be restored after perturbation. Forgetting is the loss of repair paths, not the erasure

of stored data. This connects to the deformation-of-the-admissibility-manifold account of memory loss developed separately in the *Cost of Forgetting* program.

Understanding. Understanding a distinction is possessing an efficient repair path for it. Understanding increases with the depth and economy of available repair trajectories, not with the volume of information received.

Intelligence. Intelligence, under this framework, is repair efficiency: the capacity to restore distinctions with minimal cost across the widest possible range of perturbations. The admissibility manifold of an intelligent system is larger and more robustly bounded than that of a less intelligent system.

Science. Science is collective repair under intersubjective constraint. The objectivity of scientific knowledge consists not in its correspondence to an observer-independent world but in its intersubjective restorability: the distinctions that science maintains are those restorable by the collective repair apparatus, reproducibly, across observers.

The following problems remain open.

1. **Limit restorability theorem.** Theorem 7.2 characterises $\text{Rest}_f(\mathcal{R})$ via stable observability. Prove the analogue for $\text{Rest}_\omega(\mathcal{R})$ under weaker topological assumptions: does there exist a pre-theoretic notion of *asymptotic observability* that identifies exactly with limit restorability?
2. **Interference matrix spectrum.** Characterise which symmetric real matrices arise as interference matrices $\mathbf{I}$ of some repair repertoire. What constraints does the repair structure impose on the spectrum? Is there a repair analogue of the Wigner semicircle law for large randomly generated observers?
3. **Repair conflict graph.** The positive part of $\mathbf{I}$ defines a signed graph on $\text{Rest}(\mathcal{R})$ whose positive-weight cliques are repair bottleneck clusters. Characterise graphs arising from repair repertoires. Relate the clique number to attentional capacity limits in biological observers. The empirical finding that production embedding models concentrate variance into roughly 16 effective dimensions regardless of nominal dimensionality [19] suggests that biological and artificial retrieval systems may share a characteristic in-

interference graph structure; characterising this structure in repair-theoretic terms is an open problem.

4. **Repair basis.** Is there a notion of *repair basis* for a distinction space, analogous to a basis for a vector space? A natural candidate: a minimal set $B \subseteq \text{Rest}(\mathcal{R})$ such that every element of $\text{Rest}(\mathcal{R})$ is limit-restorable from synergistic combinations of elements of B .
5. **Repair dynamics and manifold deformation.** How does A_O evolve when $\mathcal{R}_O$ is itself perturbed? Is manifold deformation under memory loss (the Cost of Forgetting program) equivalent to repair-repertoire reduction $\mathcal{R} \rightarrow \mathcal{R}' \subsetneq \mathcal{R}$? Is the resulting deformation of $\text{Rest}(\mathcal{R})$ under d_R an isometric embedding or a metric collapse?
6. **Control–Geometry Collapse without compactness.** Theorem 8.17 assumes X_S compact and convex. What are the weakest conditions under which $\text{Rest}(\mathcal{R}_O) = \text{Reach}(A_O)$? Characterise the Lie algebra rank conditions on $\mathcal{R}_O$ under which the inclusion collapses, and map this onto topological controllability of RSVP field profiles.
7. **Interface theorems.** Prove a precise connection between the Repair Boundary Theorem (Theorem 8.21) and the interface-derivation program in quadratic gravity. Under the duality of Section 8, the quantum-classical interface is the locus where quantum repair operators cease to be available at the classical substrate, i.e., where C_R diverges for quantum state-preparation distinctions. Make this precise using the Reachability–Restorability framework.

Observation is not passive reception. It is the maintenance of a world under perturbation. The world an observer inhabits is exactly as large as its capacity to repair.

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